Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 18 · Part D: Digital Signal Processing Level 3

Track: Outlier Rejection, Detrending, Savitzky-Golay, Wiener Filter, Wavelet Denoising, Gap Handling

Prerequisites: Problems 07, 14, 17 · **Estimated effort:** 14–18 hours

Denoising and Data Cleaning: Removing Noise Without Removing the Science

Every cleaning step can erase the signal you are after. On a synthetic exoplanet light curve with known truth, reject cosmic-ray outliers without clipping the transit, flatten stellar variability without eating the dip, derive the Savitzky–Golay filter, compare Wiener and wavelet denoising, and measure the *bias* each step introduces, not just the noise it removes.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/18_denoising_and_data_cleaning/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections,

and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

Raw instrument data is never ready for analysis. A space photometer's light curve contains cosmic-ray hits (isolated outliers), thermal and pointing systematics (slow trends), stellar variability, white noise and data gaps. Each can be handled, but each fix is a filter, and every filter distorts something. A median filter that is too short deletes a transit; a smoother that is too long lets the trend through; clipping the lowest points removes exactly the dips you are searching for.

The remedy is to treat cleaning like any measurement: test it on data where the truth is known, and report not only the noise reduction but also the bias in the quantities of interest (here the transit depth and timing). This problem gives you the tools and, more importantly, the habit.

Theoretical Framework

1. Robust Statistics and Outlier Rejection

The sample standard deviation is inflated by outliers; use $\hat{\sigma} = 1.4826\,\mathrm{MAD}$. *Sigma clipping* iteratively removes points beyond $n\hat{\sigma}$ of a running centre; the *Hampel filter* compares each point with the median of its neighbours and replaces it if $|x_i - \mathrm{med}| > n \cdot 1.4826\,\mathrm{MAD}$. One-sided clipping (upper outliers only) protects negative features such as transits.

2. Detrending and Its Bias

A running median or spline of window W removes variations slower than $\sim W$ and leaves faster ones. For a transit of duration T_{tr} the trend filter must satisfy $W \gtrsim 3T_{\mathrm{tr}}$, otherwise it follows the dip and suppresses its depth. A better practice is to *mask* in-transit points when fitting the trend.

3. Savitzky–Golay Filter

Fit a polynomial of degree p by least squares to each window of $2m+1$ points and take its central value (or derivative). This is an FIR filter whose coefficients are the central row of $(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ with $A_{ij} = j^i$. For $p = 2$, $2m+1 = 5$ the smoothing coefficients are $\frac{1}{35}(-3,\ 12,\ 17,\ 12,\ -3)$. It preserves polynomial moments up to order p and thus peak heights and widths better than a moving average.

4. Wiener Filter

For a signal with PSD S_s in additive noise with PSD S_n, the linear filter that minimises the mean-square error is $H(f) = S_s(f)/(S_s(f) + S_n(f))$, with minimum error $\int S_s S_n/(S_s + S_n)\,\mathrm{d}f$.

5. Wavelet Shrinkage

Take a DWT, shrink the detail coefficients with a soft threshold $\eta_\lambda(d) = \operatorname{sgn}(d)\max(|d| - \lambda, 0)$ and invert. The universal threshold is $\lambda = \hat{\sigma}\sqrt{2\ln N}$ with $\hat{\sigma} = \mathrm{MAD}(d_1)/0.6745$ from the finest-level coefficients d_1. For Poisson data the Anscombe transform $y = 2\sqrt{x + 3/8}$ stabilises the variance to about 1 so that Gaussian methods apply.

6. Gaps

Interpolating across gaps (linear, cubic, or Gaussian-process) invents data; mask gaps rather than fill them for any spectral analysis (problem 13 uses Lomb–Scargle precisely for this reason).

Data and Simulation Specification

Light curve	`sp_common.synth.transit_lightcurve(days=90, cadence_min=30, period=3.2, depth=0.01, duration_h=3.0, noise_ppm=400, variability_ppm=3000, outlier_frac=0.003, gap_frac=0.05)`; the truth masks `in_transit` and `outlier` and the clean transit model are returned.
Narrow peak	Gaussian peak of width 5 samples and unit height in noise $\sigma = 0.1$, for the Savitzky–Golay and moving-average comparison.
Wiener test	Sinusoid (20 Hz, amplitude 1) plus coloured noise with a known PSD.
Poisson data	`sp_common.synth.grb_event_list(seed=7)` binned at 0.5 s.
Real data (choose one)	A Kepler or TESS light curve from MAST via `lightkurve`: Kepler-10 (as in the session notebook) or a deep-transit host such as Kepler-5 or WASP-18 (TESS), PDCSAP flux. A noisy ECG from PhysioNet (MIT-BIH, baseline wander and mains interference) is an alternative.

Your Task

1. **Characterise the problems (Stage 2).** Plot the raw light curve, its histogram and its PSD (problem 13). Report the standard deviation and 1.4826 MAD, the number of outliers, the fraction of cadences lost to gaps and the dominant trend timescales.
2. **Outlier rejection.** Implement two-sided sigma clipping, one-sided (upper) clipping, a Hampel filter and a running-median residual test. Using the truth mask compute true- and false-positive rates against threshold $n = 2, \ldots, 8$, and the number of *in-transit* points wrongly removed by each. Recommend a method and threshold.
3. **Detrending and bias.** Flatten the light curve with a running median and a smoothing spline for window lengths $W = 0.3$ to 10 days. For each, fit the transit depth by comparing the folded in-transit and out-of-transit means and plot the depth bias $\hat{d}/d$ against W/T_{tr}; show the $W \gtrsim 3T_{\mathrm{tr}}$ rule. Repeat with transit points masked in the trend fit.
4. **Savitzky–Golay.** Derive the $(5, 2)$ smoothing coefficients by hand through the least-squares normal equations, implement the general design and compare with `scipy.signal.savgol_coeffs`. On the narrow-peak data compare peak height and width preservation of the moving average and Savitzky–Golay for several windows, and estimate a first derivative.

5. **Wiener filter.** Build the Wiener filter for the sinusoid-in-coloured-noise problem from the known spectra, apply it, and compare the mean-square error with the theoretical minimum and with an ideal band-pass filter.
6. **Wavelet denoising.** Denoise the cleaned light curve with PyWavelets: choose the wavelet, level, hard versus soft thresholding and the universal threshold. Compare RMSE, output SNR (against the clean model, `sp_common.stats.snr_db`) and transit-depth bias with Savitzky–Golay and the running median. Apply the Anscombe transform to the GRB counts, denoise and invert.
7. **Pipeline and real data.** Assemble outlier masking, trend removal with transit masking, denoising and gap flagging into a documented pipeline. Report the final noise (ppm per cadence), the depth bias and the false-positive outlier rate on the synthetic data, then run the identical pipeline on your real light curve and show the phase-folded transit with an estimated depth and uncertainty.

Self-check targets

The $(5, 2)$ Savitzky–Golay smoothing coefficients are $(-3, 12, 17, 12, -3)/35$. For $N = 1024$ and unit noise the universal wavelet threshold is $\sqrt{2 \ln 1024} = 3.72$. The Anscombe transform gives a variance of about 1 for Poisson counts above ≈ 4. A running median with $W < T_{\mathrm{tr}} = 0.125\,\mathrm{d}$ leaves essentially no transit; the depth bias approaches 1 only for $W \gtrsim 3T_{\mathrm{tr}}$ (and always with masking).

Aerospace Application Link

A star tracker's CCD images contain cosmic-ray hits that look like stars; housekeeping telemetry contains spikes from bit errors and gaps from dropouts. Describe a lightweight onboard cleaning chain (median-of-three or Hampel on thermistor and gyro data, upper clipping on pixels, gap flags) with its computational cost, and quantify using your synthetic light curve what fraction of real events (transits, here, or genuine thermal excursions) a too-aggressive filter would remove. Recommend the filter length that keeps the loss below 1%.

Deliverables

D1 **Data characterisation:** histogram, PSD, noise and outlier statistics.

D2 **Outlier study:** the four methods with true/false-positive curves and the in-transit loss.

D3 **Detrending bias:** depth bias versus window length with and without masking.

D4 **Savitzky–Golay and Wiener:** derivation, comparison with SciPy, peak-preservation study, Wiener MSE versus theory.

D5 **Wavelet denoising and pipeline:** method comparison table and the documented end-to-end pipeline with the real-data result.

D6 **Memo (250 words):** "How do we clean data without biasing the measurement?" Cite at least two external sources.

Bonus (a) Implement a biweight location filter for detrending and compare. (b) Use a Gaussian process to interpolate gaps and compare with linear interpolation. (c) Implement SURE thresholding.

Learning Resources

Denoising Theory

- Savitzky, A., Golay, M.J.E. (1964). "Smoothing and differentiation of data by simplified least squares procedures." Anal. Chem. 36, 1627.
- Donoho, D.L., Johnstone, I.M. (1994). "Ideal spatial adaptation by wavelet shrinkage." Biometrika 81, 425.
- Mallat, S. *A Wavelet Tour of Signal Processing*, chapter on denoising by thresholding.
- Kay, S.M. *Fundamentals of Statistical Signal Processing, Vol. III: Practical Algorithm Development* and Vol. I (Wiener filtering).

Astronomy and Light Curves

- Borucki, W.J. et al. (2010). "Kepler planet-detection mission: introduction and first results." Science 327, 977.
- Lightkurve documentation (flattening, outlier removal, folding): `https://lightkurve.github.io/lightkurve/`; MAST archive: `https://archive.stsci.edu`
- Anscombe, F.J. (1948). "The transformation of Poisson, binomial and negative-binomial data." Biometrika 35, 246.

Tools

- SciPy `signal.savgol_filter`, `signal.medfilt`; PyWavelets: `https://pywavelets.readthedocs.io`; PhysioNet MIT-BIH: `https://physionet.org/content/mitdb/1.0.0/`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.